

UI for Free Fall Penetrometers – Project Charter

–

1. Project

§1.1 – Project Vision

- ❖ To provide a simple, unified interface for data derived from free fall penetrometers to be entered, analyzed, and visualized in order to make predictions for the type of soil.

§1.2 – Success Criteria

- ❖ Up to 300 binary data files from free fall penetrometers can be entered into the system to plot acceleration over time
- ❖ Individual penetrometers are mapped to constants to be used in calculations so that a user does not have to manually edit the code to change the constants
- ❖ A singular desktop application combines the current functionality that is spread across Matlab, Python, and Excel

§1.3 – MOV

- ❖ Within one academic year, a desktop application is created that combines the existing Matlab, Python, and Excel functionality into one system that takes a user less than 1 hour to generate visuals that can be analyzed to determine soil type from 8 hours worth of penetrometer data.

2. Team

§2.1 – Stakeholders

- ❖ Product Owner: Julie Paprocki
- ❖ Scrum Master: David Benedetto
- ❖ Development Team:
 - Austin Snow
 - acs1114@usnh.edu
 - Brandon Kady
 - btk1015@usnh.edu
 - Kabir Bardai
 - kab1206@usnh.edu
 - Boyd Crapo
 - brc1051@usnh.edu
 - Connor Harrington
 - cah1099@usnh.edu

§2.2 – Responsibilities

- ❖ The development team will meet weekly in Kingsbury Hall with the Product owner on Wednesdays at 12:40 PM.
- ❖ The development team is also responsible to meet for stand ups at 12:00 PM on Fridays with an update on their current task
- ❖ The project owner will need to aid the development team in understanding the current functionality and determining the priorities of additional functionality

3. Scope

§3.1 – Project Risks

- ❖ The compatibility of the existing systems
- ❖ The speed of data visualization in Python compared to Matlab
- ❖ The size of the data from one day of collection may exceed the input size for some data analysis libraries

§3.2 – Project Size and Complexity

- ❖ Previous code will need to be reviewed and reformatted to work together in one system
- ❖ A user interface will need to be created that will be easy to understand and use.
- ❖ The system will not be responsible for generating the likelihood of each soil type, the user will be responsible for interpreting the data

4. Roadmap

§4.1 – MVP

- ❖ The desktop application combines the functionality of previously utilized scripts, allowing simple interaction to provide visualization of the data.

§4.2 – Epics

- ❖ Unifying Existing Systems
 - Combine existing Matlab, Excel, and Python scripts into one unified system
 - Ensure that no changes to the code needs to be made, values that need to be variable are given by user input
- ❖ Establishing User Interface
 - Create a graphical desktop application to increase the efficiency of entering and processing binary data from the penetrometers.

§4.2.1 – Epic Timeline

